ASSESSMENT TOOL -2

CO1-AT2-CONCEPT MAPPING QUESTIONS

Course Code: ITA0609

Course Title: Machine Learning

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CO1-AT2 Concept Mapping Assignment

Question 1

1. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.Introduction

Introduction

Machine learning can analyze patient records to predict diseases early and support doctors in diagnosis.

Problem Statement

Predict whether a patient has heart disease or diabetes using age, blood pressure, glucose level and lifestyle factors.

Suitable Algorithm

Random Forest Classifier.

Justification

It provides high accuracy, handles numerical and categorical data, reduces overfitting and identifies important features.

Data Preprocessing

Collect data, remove missing values, encode categorical values, normalize features and split into training/testing datasets.

Feature Selection

Select important features such as age, blood pressure, glucose level, BMI, smoking, exercise and family history.

Model Training

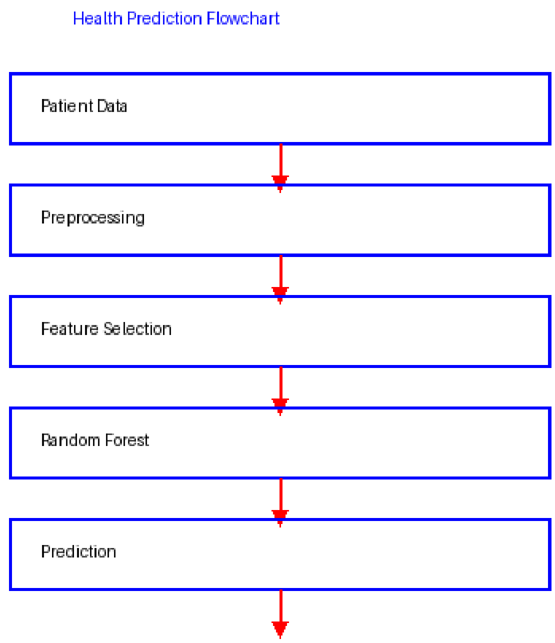
Train the Random Forest model using training data, validate using test data and evaluate with Accuracy, Precision, Recall and F1-score.

Expected Output

Predict Disease = Yes/No with high accuracy.

Conclusion

The proposed system helps early disease prediction and improves healthcare decision making.



Accuracy=(TP+TN)/(TP+TN+FP+FN)
Example: TP=90 TN=85 FP=5 FN=10 => Accuracy=92.1

Question 2

In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.

SOLUTION :

Ex	Employment	Salary	Credit	History	Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

Positive Examples

Examples 1, 2 and 4 are positive because Loan Approved = Yes.

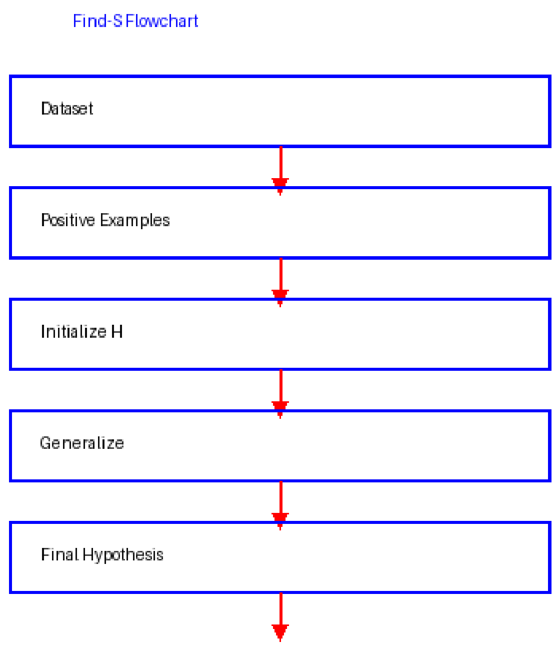
Find-S Steps

Initial Hypothesis: <Ø, Ø, Ø, Ø>
After Example 1: (Employed, High, Good, Clean)
After Example 2: (Employed, ?, Good, Clean)
After Example 4: (Employed, ?, ?, Clean)

Final Hypothesis

H = (Employed, ?, ?, Clean)

Conclusion: A loan is approved for applicants who are Employed and have a Clean loan history, regardless of Salary and Credit Score.


Find-S Flowchart
Dataset
Positive Examples
Initialize H
Generalize
Final Hypothesis